2.2 What Is Osmosis?

Biology · Movement of Substances · Syllabus refs (G2 & G3): 2b

Name: _____ Class: _____ Date: _____

Learning outcomes

- Define osmosis, investigate and describe the effects of osmosis on plant and animal tissues.
- Discuss the factors that affect the rate of osmosis.

Osmosis

The cell membrane is a **partially permeable membrane**: it lets some substances through but not others. In Fig. 2.8, a partially permeable membrane separates the two arms of a U-tube. The membrane lets **water** molecules through, but **not sucrose** molecules. (Sucrose is cane sugar, the sugar you use at home.)

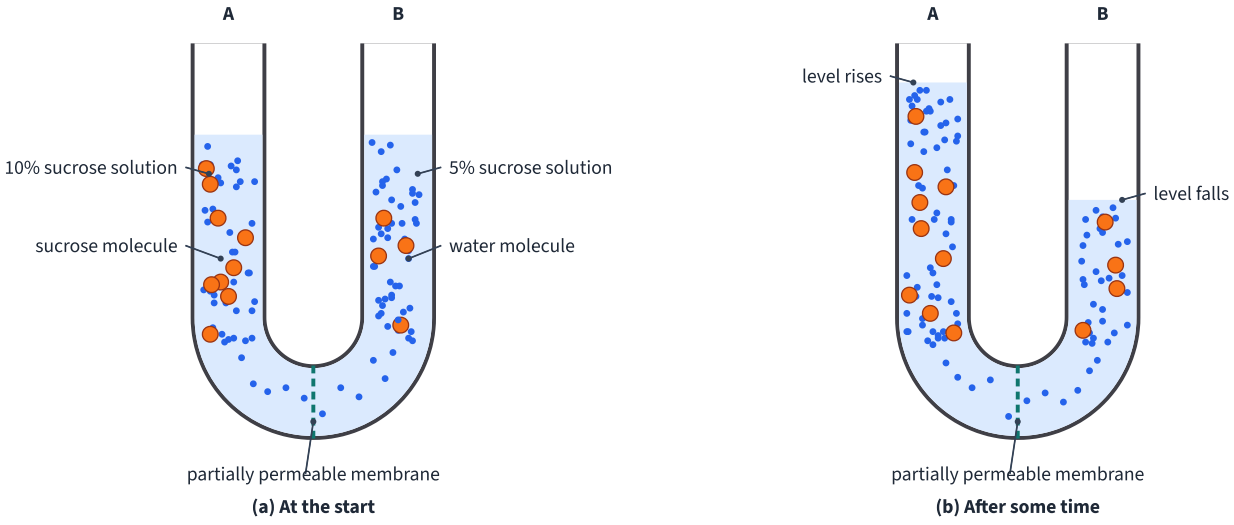


Fig. 2.8 — Experiment to show osmosis.

Compared with arm B, arm A has:

- **more** sucrose molecules per unit volume;
- **fewer** water molecules per unit volume.

After some time, the level in arm A **rises** and the level in arm B **falls**. Why?

- Sucrose molecules are too large to pass through the tiny pores of the membrane, so they cannot move from A to B.
- Water molecules move from B to A, because solution B has more water molecules per unit volume than solution A. (A **dilute** solution has more water molecules per unit volume than a **concentrated** solution.)

Water keeps moving until both arms have the same solute concentration (the same number of solute molecules per unit volume).

> The diffusion of **water molecules** across a **partially permeable membrane** is called **osmosis**.

> **Link to Chemistry**
>
> The **concentration** of a solution is the amount of solute dissolved in a unit volume of the solution.

Practical: demonstrating osmosis

Aim: to show osmosis using dialysis tubing (Visking tubing), which is a partially permeable membrane.

Procedure

1. Tie a knot in one end of a piece of dialysis tubing.
2. Fill a boiling tube halfway with distilled water.
3. Use a pipette to fill the dialysis tubing halfway with sucrose solution. If any spills, rinse the outside of the tubing with distilled water.
4. Set up the apparatus as in Fig. 2.9. The liquid levels inside and outside the tubing must be the same. Hold the tubing in place with a paper clip and mark the level on the outside of the boiling tube.
5. Set up a second boiling tube in the same way, but with distilled water both inside the tubing and in the boiling tube.

6. After 20 minutes, observe the liquid levels inside and outside the tubing in each set-up.

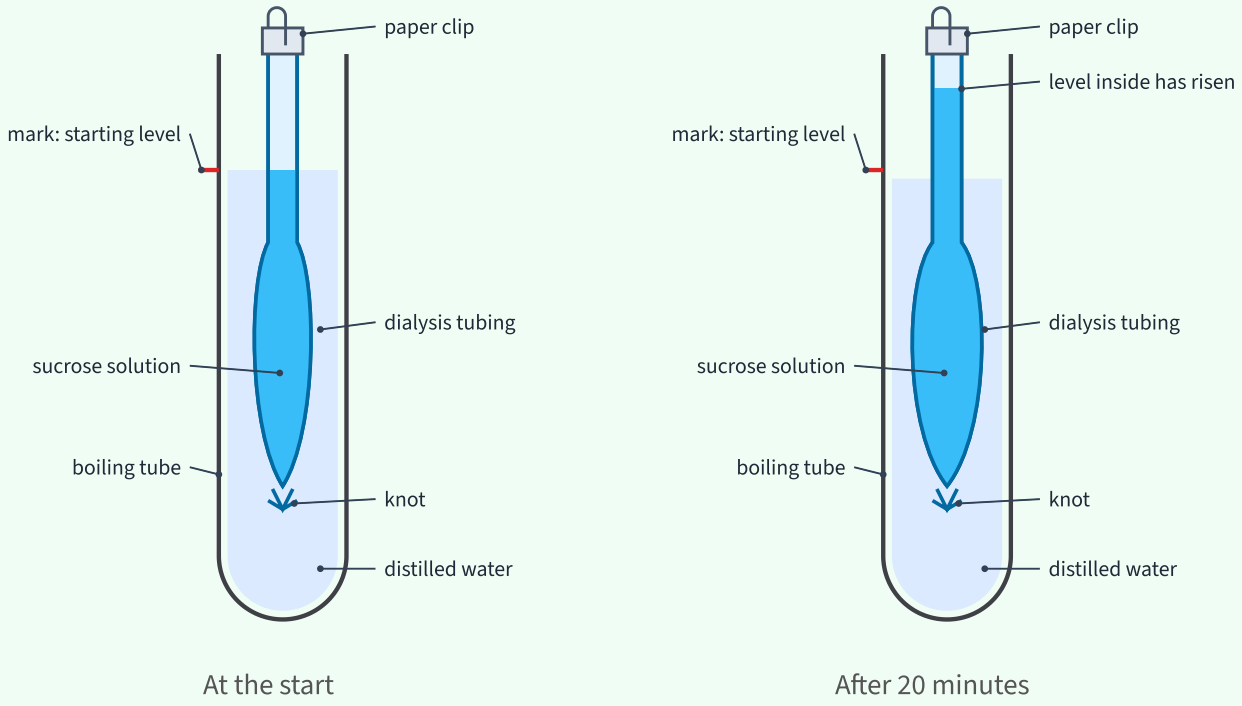


Fig. 2.9 — Set-up to demonstrate osmosis.

Observation and discussion

1. What do you notice about the dialysis tubing after 20 minutes? Explain your observation.
2. What is the purpose of the second set-up (distilled water inside and outside)?

What is water potential?

When we talk about osmosis we use the term **water potential**. Water potential is a measure of how likely water molecules are to move from one place to another.

- Water molecules tend to move from a **dilute** solution (less solute) to a **more concentrated** solution.
- So a **dilute** solution has a **higher** water potential than a concentrated solution. Pure (distilled) water has the highest water potential of all.

- When a partially permeable membrane separates two solutions with different water potentials, a **water potential gradient** is set up. Water molecules move from the region of higher water potential to the region of lower water potential: **down** the water potential gradient.

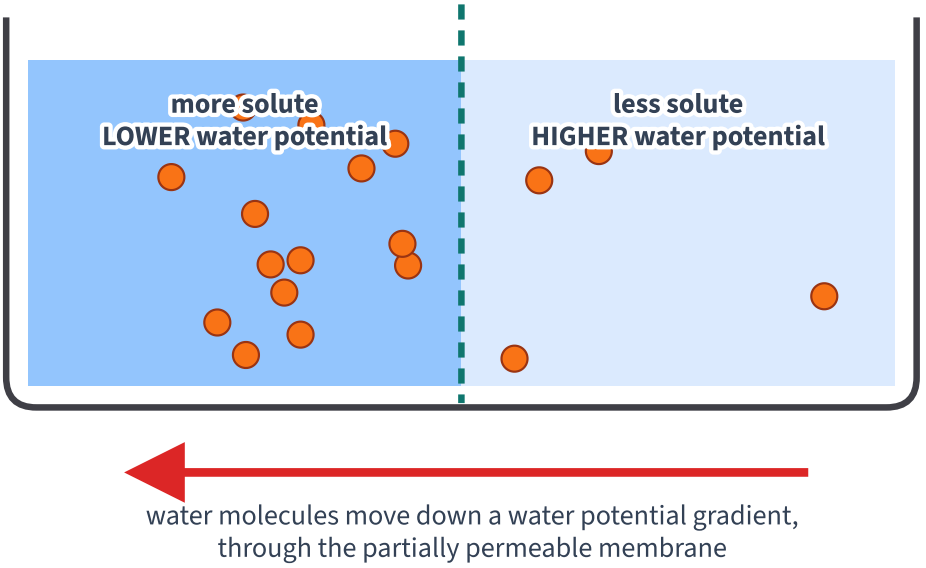


Fig. 2.10 — A water potential gradient across a partially permeable membrane.

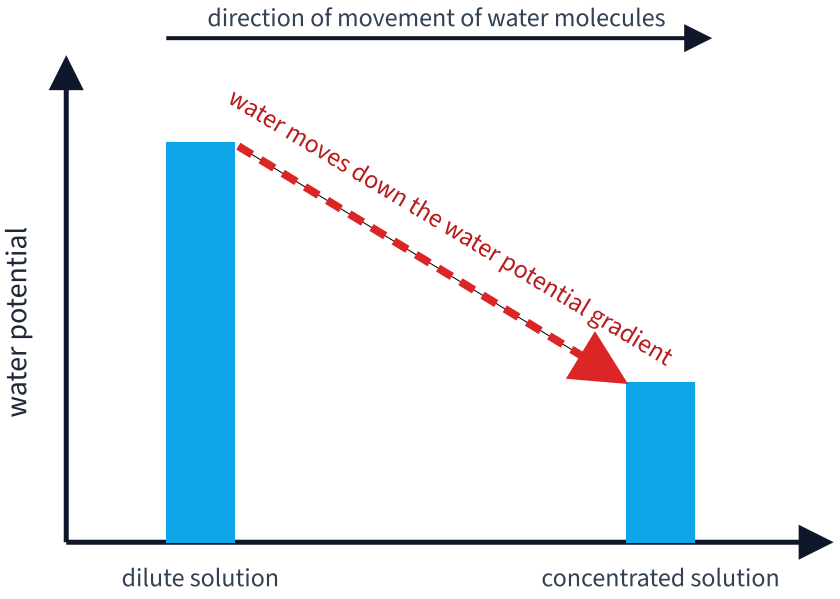


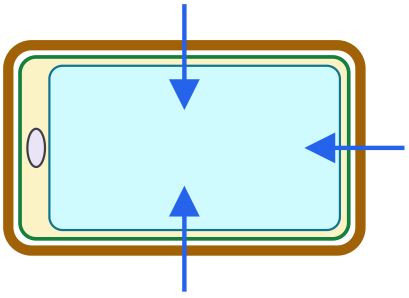
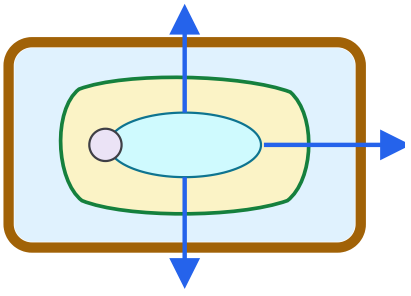
Fig. 2.11 — Water moves from a dilute solution (higher water potential) to a concentrated solution (lower water potential).

Osmosis is the net movement of water molecules from a region of higher water potential to a region of lower water potential, through a partially permeable membrane.

Common mistakes: in osmosis it is the *water* that moves, not the solute. More solute means *lower* water potential. Always mention the *partially permeable membrane* in your answer.

What happens to plant cells in different solutions?

The cell sap in the vacuole of a plant cell is a solution. Whether water goes in or out depends on the water potential of the solution outside the cell.

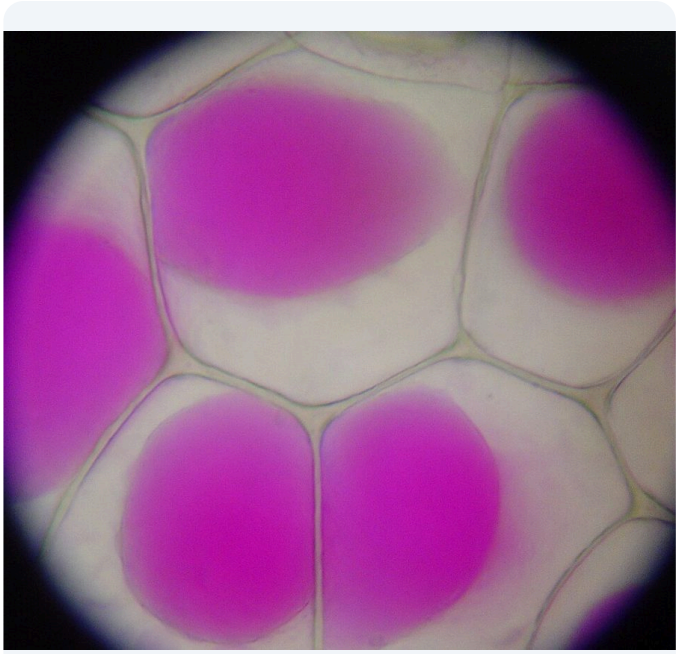
	Plant cell in a solution of higher water potential (e.g. distilled water)	**Plant cell in a solution of lower water potential (e.g. concentrated sugar solution)**
Water potential of cell sap compared with the solution outside	Lower than the solution outside.	Higher than the solution outside.
Movement of water by osmosis	Water molecules **enter** the cell through the partially permeable cell membrane.	Water molecules **leave** the vacuole and cytoplasm through the partially permeable cell membrane.
What happens to the cell	The vacuole gets bigger and pushes the cytoplasm against the cell wall. The cell does **not burst**, because the strong, inelastic cell wall protects it. The pressure of the water in the vacuole is the **turgor pressure**. 	The vacuole gets smaller. The cytoplasm shrinks and the cell membrane pulls away from the cell wall. This is called **plasmolysis**; the cell is **plasmolysed**. 
Result	The cell swells and becomes **turgid** (its firmness is called **turgor**).	The cell shrinks and becomes **flaccid** (limp).

Blue arrows: the direction water moves by osmosis. Cell drawings not to scale.

A plasmolysed cell can recover. If it is put back into water (or a solution of higher water potential) soon enough, water enters by osmosis and the cell becomes turgid again.



Red onion cells in a concentrated salt solution. Look for cells where the coloured contents have shrunk away from the cell wall: these cells are **plasmolysed**.

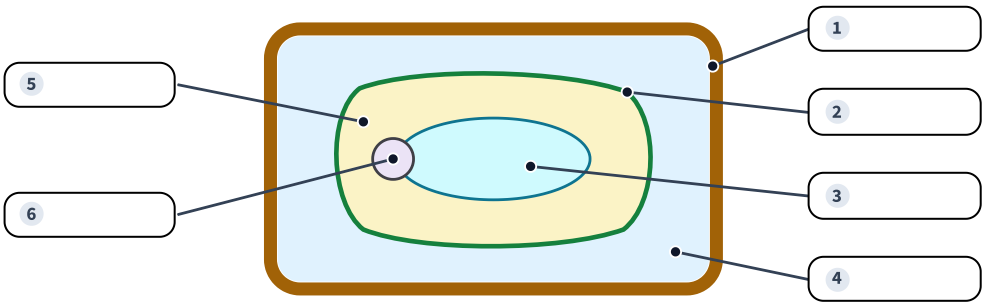


Close-up of plasmolysed cells. The pink vacuole has shrunk and the cell membrane has pulled away from the cell wall. The clear gaps are filled with the outside solution.

Images: Trương Minh Khải, CC BY 4.0; Mnolf, CC BY-SA 3.0; via Wikimedia Commons.

Label a plasmolysed plant cell.

nucleus | cytoplasm | outside solution | vacuole | cell membrane | cell wall

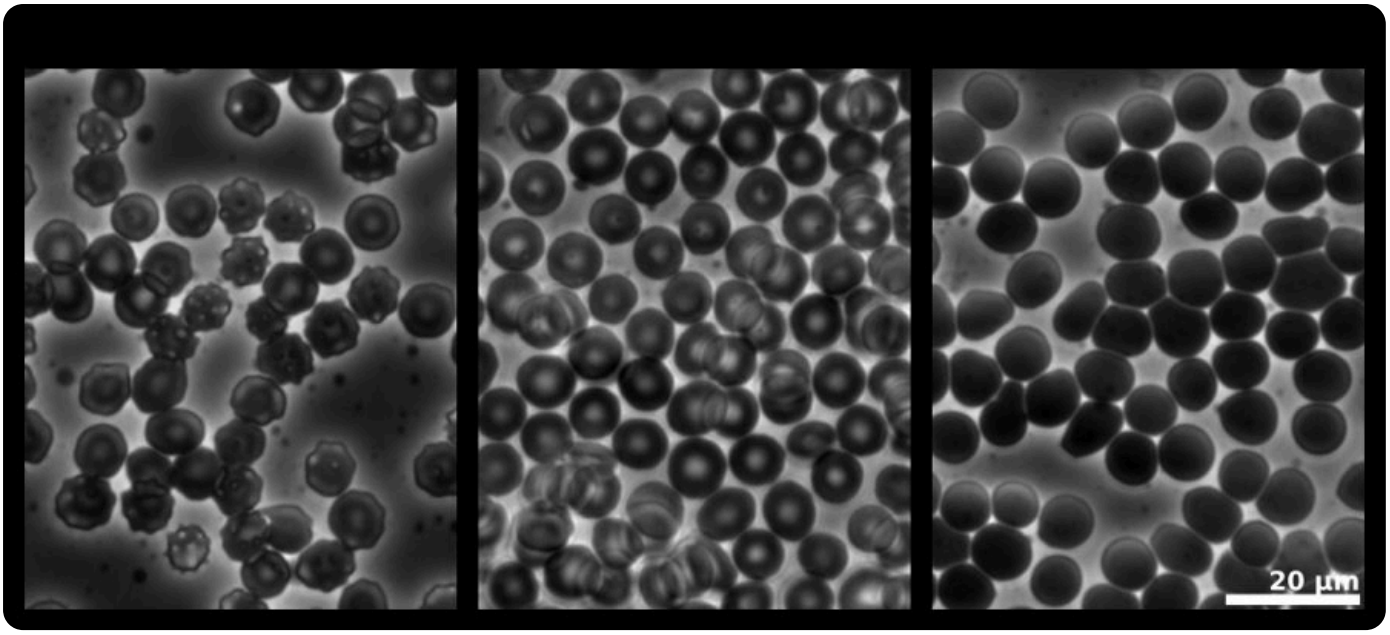


A plasmolysed plant cell.

What happens to animal cells in different solutions?

	Animal cell in a solution of higher water potential	Animal cell in a solution of lower water potential
Water potential of cytoplasm compared with the solution outside	Lower than the solution outside.	Higher than the solution outside.
Movement of water by osmosis	Water molecules move **into** the cell.	Water molecules move **out of** the cell.
What happens to the cell	The cell swells and may **burst**. Unlike a plant cell, it has no cell wall to protect it.  normal → swells → bursts	The cell shrinks and spikes appear on its surface. This is called **crenation**. The cell becomes dehydrated and will die if it stays there.  normal → crenated (shrunken, with spikes)

In a solution with the same water potential as its cytoplasm, a cell does not change size or shape. Water molecules move in and out at the same rate, so there is **no net movement** of water. This is why medical drips and eye drops are made with the same water potential as body fluids.

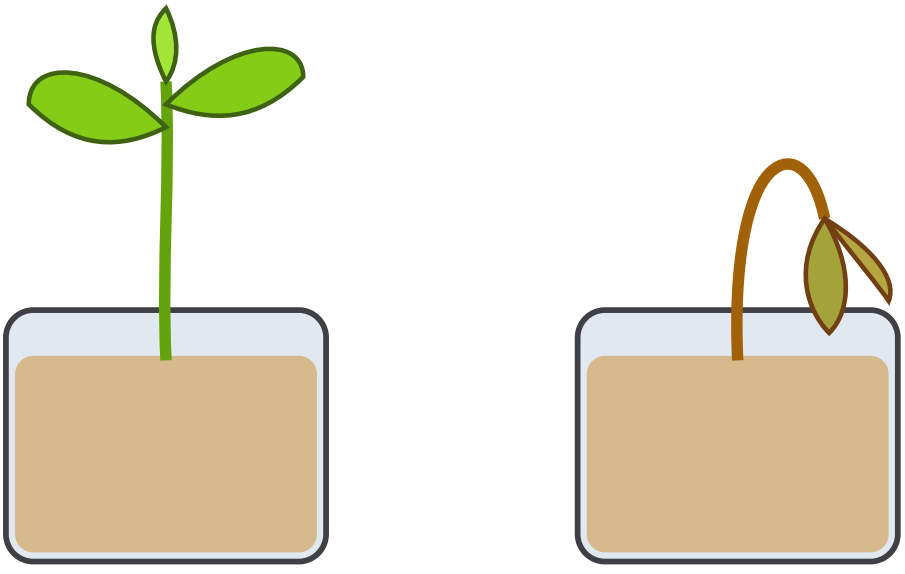


Real red blood cells (light micrographs). **Left:** in a solution of lower water potential: crenated, with spikes. **Middle:** in a solution of the same water potential: normal discs. **Right:** in a solution of higher water potential: swollen and round.

Image: Zephyris, CC BY-SA 3.0, via Wikimedia Commons.

Why is turgor important to plants?

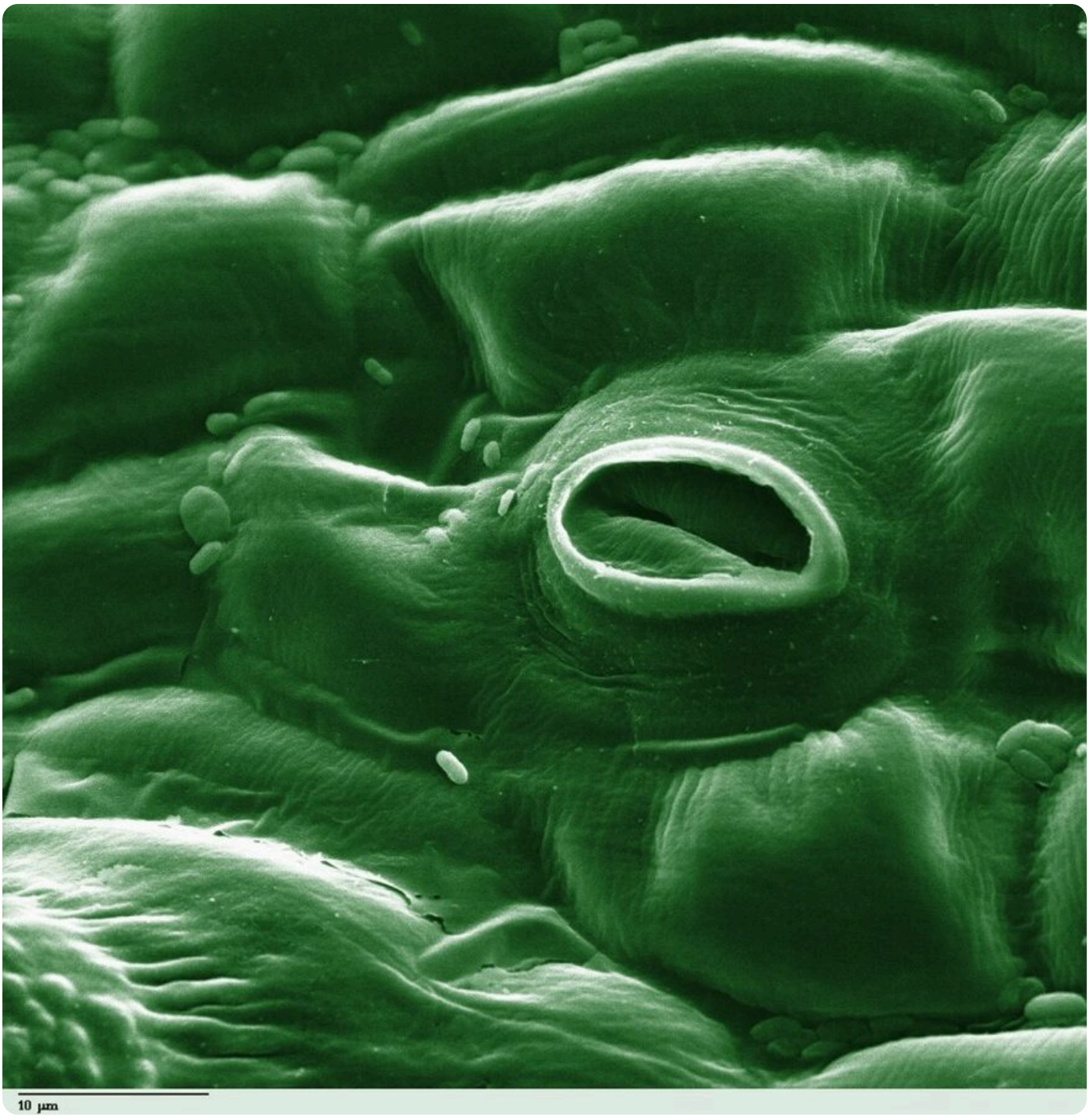
Support. Turgor keeps soft plant parts firm. Young stems and the leaves of non-woody plants stay firm and upright because of the turgor pressure in their cells. If the cells lose too much water, they lose turgor, become flaccid, and the plant **wilts**.



(a) Turgid: firm and upright **(b) Lost turgor: wilting**

Fig. 2.12 — Turgor pressure keeps a young plant upright; losing turgor makes it wilt.

Movement. Changes in turgor make some plant parts move. For example, a stoma (a tiny pore in a leaf) opens and closes when its two **guard cells** gain or lose turgor.



A stoma on a tomato leaf (scanning electron micrograph, colour added). When the two guard cells around the pore become turgid, the stoma opens.

Image: Photohound, public domain, via Wikimedia Commons.

Too much fertiliser. Plasmolysed cells die if they stay plasmolysed for too long. If too much fertiliser is added around a plant, the **soil solution** (the thin layer of water around soil particles, with dissolved mineral salts) becomes very concentrated. Its water potential becomes lower than that of the root cells, so water moves **out** of the roots by osmosis. The roots cannot absorb

water while the leaves keep losing water by evaporation, so the plant wilts and will die unless the soil solution is diluted with water.

Factors that affect the rate of osmosis

Osmosis is the diffusion of water, so it is affected by the same kinds of factors as diffusion:

- **Water potential gradient:** the bigger the difference in water potential across the membrane, the faster the rate of osmosis.
- **Distance:** the shorter the distance the water molecules must travel, the faster the rate.
- **Surface area to volume ratio:** the larger the SA : V of the cell or tissue, the faster water moves in or out.

Practical: effect of osmosis on plant tissue

1. Cut potato strips of the same size (for example 5 cm long), without skin. Measure the length (or mass) of each strip.
2. Put one strip into each of: distilled water, a dilute sucrose solution and a concentrated sucrose solution.
3. After 30 minutes, take the strips out, blot them dry and measure them again. Feel how firm or soft they are.

Results: in distilled water the strip gets **longer** and firm (water enters the cells by osmosis; the cells become turgid). In the concentrated solution it gets **shorter** and soft (water leaves the cells by osmosis; the cells become flaccid or plasmolysed). Work out the **percentage change** so strips of different starting sizes can be compared.

> **percentage change = (final length − starting length) ÷ starting length × 100%**

A. A potato strip was 5.0 cm long. After 30 minutes in a concentrated sucrose solution it was 4.6 cm long. Work out the percentage change in length (use a minus sign for a decrease).

Your answer (%)

Check your understanding

1. Which is the correct definition of osmosis?

- ○ The net movement of solute molecules from a region of higher water potential to a region of lower water potential
- ○ The net movement of water molecules from a region of higher water potential to a region of lower water potential, through a partially permeable membrane
- ○ The net movement of water molecules from a region of lower water potential to a region of higher water potential, through a partially permeable membrane
- ○ The movement of any particles through a fully permeable membrane

2. Which solution has the highest water potential?

- ○ Distilled water
- ○ 5% sucrose solution
- ○ 10% sucrose solution
- ○ 20% sucrose solution

3. In the U-tube experiment (Fig. 2.8), why does the level in arm A rise?

- ○ Sucrose moves from B to A.
- ○ Water moves from A to B.
- ○ Water moves from B to A by osmosis.
- ○ Sucrose molecules grow bigger.

4. A plant cell is placed in distilled water. Why does it not burst?

- ○ Its vacuole shrinks.
- ○ Its strong, inelastic cell wall stops it from bursting.
- ○ Water cannot enter a plant cell.
- ○ Its cell membrane is fully permeable.

5. What is plasmolysis?

- ○ When a red blood cell bursts
- ○ When the cytoplasm shrinks and the cell membrane pulls away from the cell wall
- ○ When a plant cell swells and becomes turgid
- ○ When spikes form on a red blood cell

6. Red blood cells are put into a concentrated salt solution. What will happen?

- ○ They swell and burst.
- ○ They become turgid.
- ○ They shrink and become crenated.
- ○ Nothing.

7. A farmer adds far too much fertiliser to the soil. Why might the crop wilt?

- ○ The soil solution has a lower water potential than the root cells, so water leaves the roots by osmosis.
- ○ The fertiliser blocks the stomata.
- ○ The soil solution has a higher water potential than the root cells.
- ○ The roots take in too much water.

8. In the dialysis tubing practical, why is a second set-up with distilled water inside and outside the tubing used?

- ○ To make the experiment faster
- ○ As a control, to show that the change is caused by the sucrose solution
- ○ To kill bacteria
- ○ To measure the temperature